

Team Project Proposal

BB-8: Omnidirectional Spherical Robot with Magnetic Head and Onboard Camera

CS-358 — Making Intelligent Things

EPFL Spring 2026

Team Sphere

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1. High-Level Project Description

1.1 Overview

BB-8 is an omnidirectional spherical robot inspired by the Star Wars character.

The robot consists of a 3D-printed spherical shell (35 cm in diameter) that can roll freely on a flat surface. Inside the sphere, three omnidirectional wheels are arranged symmetrically at 120° in the horizontal plane. Each wheel is driven by a 12 V DC motor, allowing the robot to generate movement vectors in any direction without requiring prior reorientation. This configuration enables both translational and rotational motion through differential wheel control.

A central vertical shaft maintains alignment between the internal drive system and the outer shell. The dome-shaped head is magnetically coupled to the top of the sphere via a ring of magnets. This magnetic coupling allows the head to remain upright while the body rotates beneath it, creating the illusion of independent motion.

The system is controlled wirelessly via a mobile device or a gamepad. User inputs are processed by a microcontroller, which computes the required wheel velocities.

Additionally, the head contains a camera module capable of live video streaming, enhancing interactivity and enabling remote operation.

Drive

BB-8 can roll in any direction thanks to a robot housed inside, running like a hamster in its ball. It can take direction via WiFi or follow a pre-programmed path.

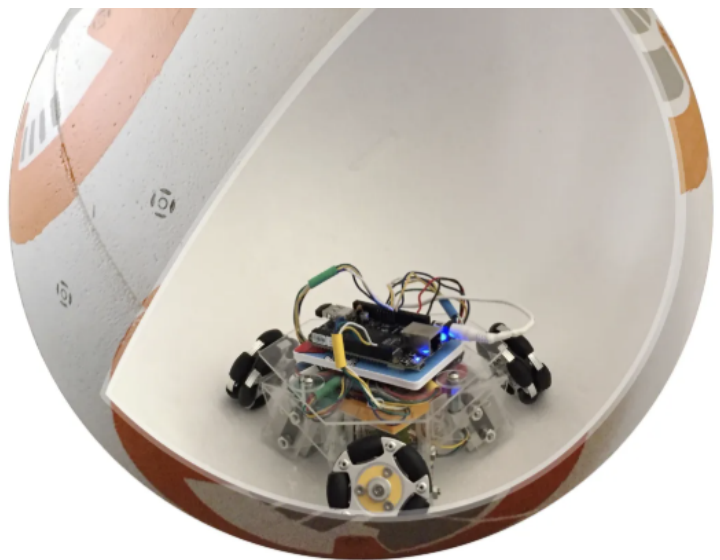


Figure 1: Internal drive platform reference.¹

¹<https://makezine.com/projects/build-full-size-working-bb-8-using-beaglebone/>

1.2 Hardware Components

- **XIAO ESP32S3 Sense**

The XIAO ESP32S3 Sense (pre-soldered) is an all-in-one module integrating a camera, a CV chip, and a microcontroller in a single compact unit (estimated 20 g). Built around the ESP32-S3 chip, it handles video streaming via WiFi, eliminating the need for an external processor.

- **BNO055 Intelligent 9-Axis Sensor**

A 9-axis IMU providing absolute orientation through sensor fusion of accelerometer, gyroscope, and magnetometer data. Chosen over a simpler 6-axis IMU because the integrated magnetometer allows drift-free absolute orientation estimates, which is critical for the stabilisation of the internal platform.

- **DC 12 V 251 RPM 18 kg·cm Motors ($\times 3$)**

Brushed DC motors driving the three omni-wheels arranged at 120° inside the sphere. The high torque (18 kg·cm) is necessary to move the full internal chassis by friction against the sphere wall, and the 251 RPM speed provides a good balance between manoeuvrability and control stability. The 12 V rating is directly compatible with a 3S LiPo battery, avoiding an additional voltage converter for the motors.

- **Double BTS7960 43 A H-Bridge Motor Drivers ($\times 3$)**

High-current motor drivers, one per motor. The 43 A rating provides a large safety margin during motor startup peaks and mechanical perturbations. The integrated H-bridge enables bidirectional control via Pulse Width Modulation (PWM) signals from the microcontroller, and one dedicated driver per motor allows fully independent speed and direction control of each omni-wheel, which is essential for holonomic locomotion.

- **LiPo Battery 3S 11.1 V 5000 mAh**

Main power source for the robot. The 11.1 V nominal voltage is directly compatible with the 12 V DC motors, avoiding an additional voltage converter. The 5000 mAh capacity provides sufficient runtime for the motors and electronics during operation, while keeping the weight and size of the battery moderate, which is important for the balance of the internal chassis.

- **Battery protection board**

The 3S 60A 12V LiPo balanced protection board safeguards the main body battery against overcharge, over-discharge, and short circuits. It supports up to 60A continuous current and includes a balancing function to maintain equal charge across all three cells of the 3S LiPo pack. This part should normally come from the CS-358 stock.

- **XT90 connectors**

The two XT90 male connectors and two XT90 female connectors to plug to the Lipo and solder to the battery management system. They allow for efficient disconnection and reconnection of the cables if we need to change the system or in case of an emergency

problem. XT90 can provide up to 90 amperes, which is more than enough for our BB8, and also fits our high-capacity (5000mAh) LiPo.

- **12 AWG cable (4mm²)**

These electric cables allow continuous currents of 20A with bursts of 25A. Our BB8's maximum amperes is 22A, when all 3 wheels and chip are consuming at the theoretical maximum, but this will not happen continuously. This cable therefore provides safety, reduces the energy loss in heating, and lets the right amount of current flow efficiently.

- **Buck Converter (7–20 V to 5 V 3 A)**

Converts the LiPo battery voltage down to a stable 5 V to power the logic-level components (microcontroller, IMU, camera). Ensures safe and regulated power delivery across mixed-voltage components without requiring a separate battery for the electronics.

- **Neodymium Disc Magnets (×6, 20 mm diameter, 5 mm thick)**

Create the contactless link between the internal shaft and the dome head. Three magnets are fixed to the top of the central shaft, three are mounted in the head, with opposing poles facing each other. This arrangement locks the head's orientation to the body while allowing it to rotate freely relative to the rolling shell, without any physical connection that would constrain the sphere's motion.

- **Li-Ion Battery 18650 3.7 V 3200 mAh (Head)**

Dedicated power source for the head. Since there is no wired connection between the head and the body, an independent battery is necessary to power the camera and servo throughout operation.

- **1×18650 Battery Shield 5 V 3 A / 3 V 1 A**

Manages charging and power delivery for the head battery. Allows recharging via USB and provides regulated 5 V and 3 V outputs, simplifying power management within the head without requiring a separate charging circuit.

- **Ball Transfer Units**

The Ball Transfer Unit is an omnidirectional rolling element that supports the head on the outer surface of the sphere, allowing it to glide smoothly in any direction with minimal friction. Rather than buying a prefabricated unit, we will build this system ourselves using steel balls

- **Wemos D1 R32**

ESP32-based microcontroller provided by the course. It manages the entire system: receiving WiFi and Bluetooth commands, reading the IMU, computing the three omni-wheel speeds, and sending PWM signals to the motor drivers.

- **SG90 Servo**

Lightweight micro-servo integrated in the head, used to adjust the tilt angle of the camera. Its small form factor and low weight make it well suited to the space constraints

of the head assembly.

- **Omni-wheels ($\times 3$, body)**

Three omnidirectional wheels positioned at 120° inside the sphere. Rather than buying prefabricated units, we will design and 3D-print the wheels ourselves. This allows us to adapt the geometry precisely to the internal dimensions of the sphere while enabling the holonomic locomotion required to move it in any direction.

All components are interconnected according to this circuit diagram:

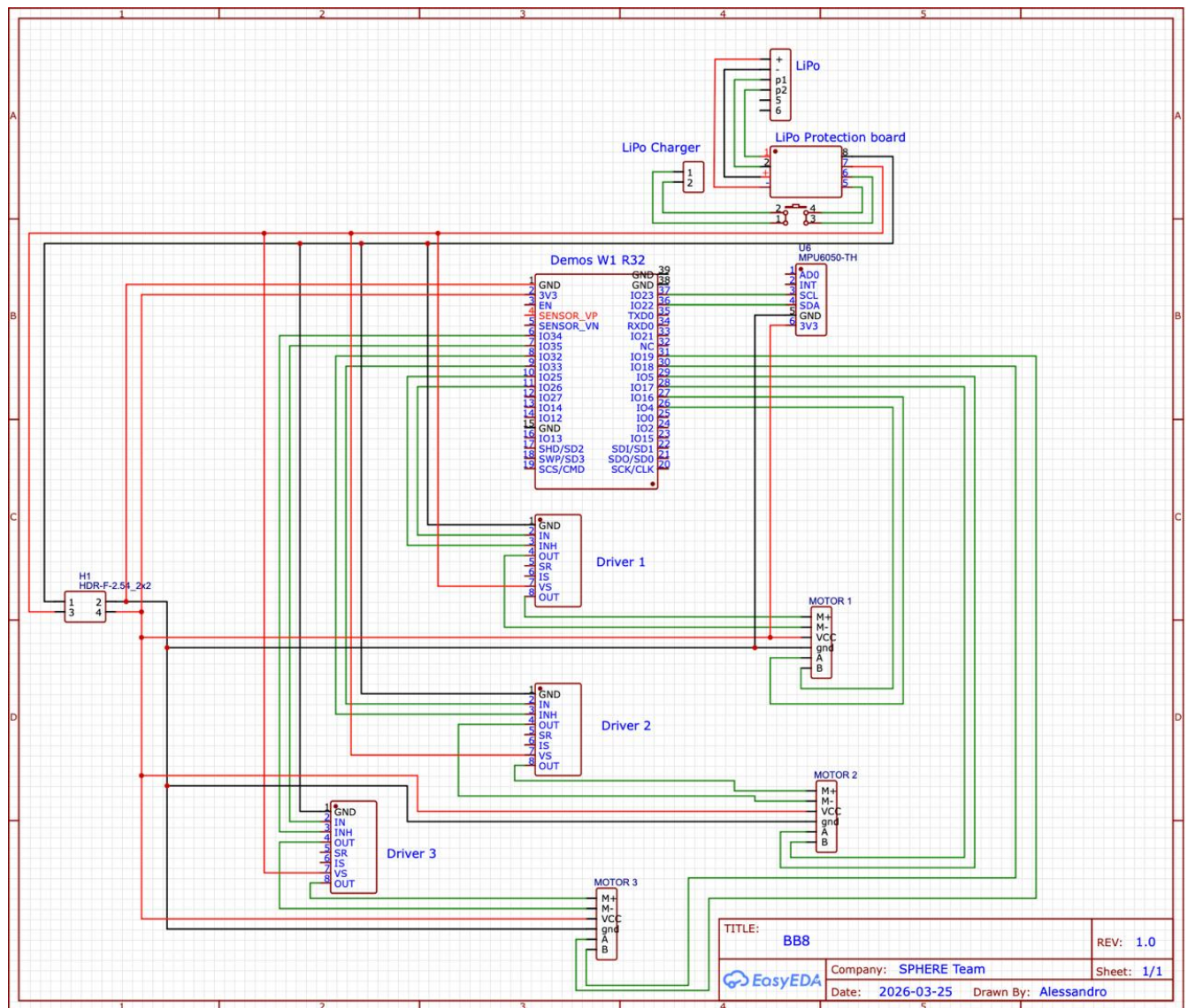


Figure 2: Circuit body

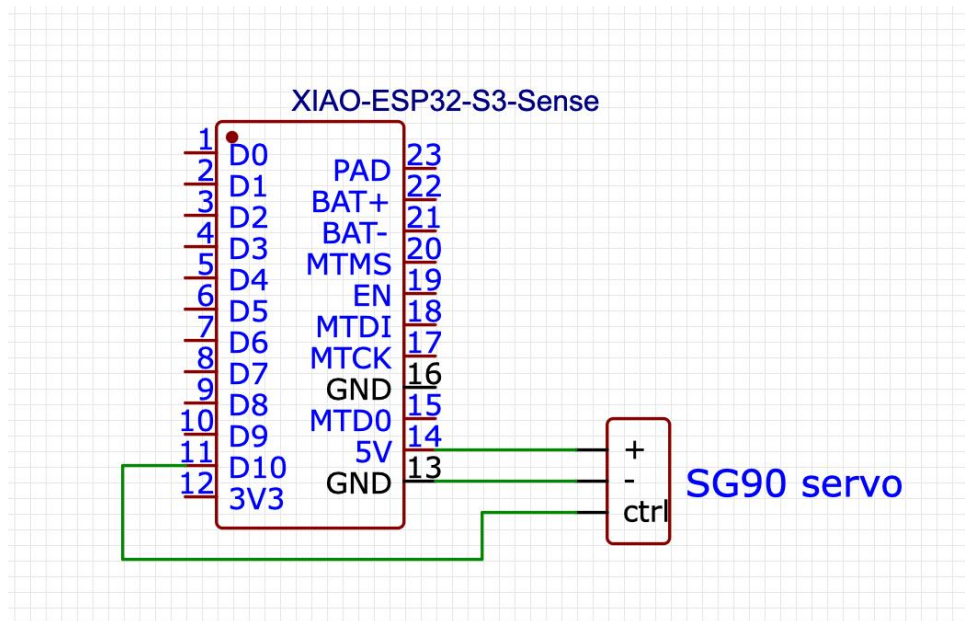


Figure 3: Circuit head

1.3 Software Architecture

1.3.1 Overview

The system is split into three software components: head firmware, body firmware, and a laptop webapp. All three communicate over WiFi on the same local network. The laptop acts as the central orchestrator in both operating modes.

1.3.2 Head firmware

The head firmware is built on the Arduino framework using the ESP32 Camera Driver. It runs two HTTP services: an MJPEG stream endpoint that continuously serves JPEG frames from the camera, and a control endpoint that accepts GET requests to set the servo tilt angle. Both run on the same HTTP server. The firmware is otherwise stateless. It does not make decisions or initiate communication. It responds to requests.

1.3.3 Body firmware

The body firmware runs a WebSocket server and a PID control loop. The WebSocket server listens for drive commands (heading, speed) from the laptop and reports telemetry (orientation, battery level) back. The PID loop runs independently at a fixed frequency, reading the IMU and computing motor outputs to maintain balance. Drive commands from the WebSocket are treated as setpoints for the PID loop, not as direct motor instructions. If no commands arrive, the loop holds the current position.

1.3.4 Webapp

The webapp runs in the browser on the operator's laptop. It connects to the head's MJPEG stream via an `<img>` tag pointing at the stream URL, and to the body's WebSocket server for sending commands and receiving telemetry. In RC mode, keyboard or joystick input is mapped to drive commands and sent over the WebSocket. Servo tilt is controlled via

HTTP GET requests to the head. In autonomous mode, the webapp captures frames from the video stream and passes them to a CV pipeline (OpenCV or YOLO running locally on the laptop). The pipeline outputs target coordinates, which the webapp converts into drive commands and sends to the body over the same WebSocket. The operator retains a kill switch in both modes.

1.3.5 Design rationale

Computer vision was originally planned to run on-device using a dedicated NPU (Grove Vision AI V2). This was abandoned after discovering that the camera module paired with it could not stream video over WiFi, only inference metadata. Moving CV to the laptop simplified the head firmware to a single responsibility (stream video, accept tilt commands) and gave the CV pipeline access to significantly more compute.

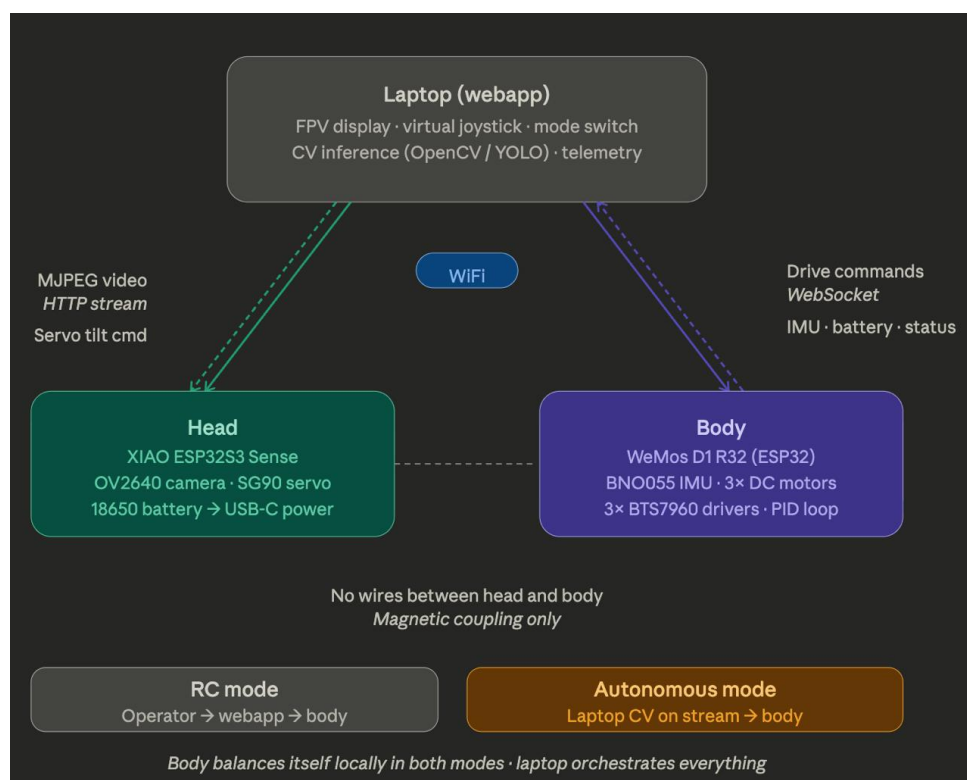


Figure 4: General Architecture

2. Related Projects

- **BB8 Builders Club** — A club of Star Wars fans that try to make replicas of the BB8 robot. They design special 3D printed pieces to assemble the spherical body, and try to make the robot as close as possible to the real BB8 in Star Wars.

Short video: https://youtu.be/aaxjrVZ7il0?si=t2LZDEK-T_tUv0oW

Facebook group (no official website found): <https://www.facebook.com/groups/BB8BuildersClub/>

- **James Bruton** — An open-source BB8. James Bruton gives valuable insight into internal mechanics, drive kinematics, electronics, ... He dives into each version of his BB8, explaining what he missed and did wrong, how he improved them and what he still has to improve.

His BB8 v3 on YouTube : https://youtu.be/J5MxFm6p5Uc?si=qsJnD_mx3jAF-TNe

- **BB8 with omnidirectional wheels** — Eric Boehlke made a BB8 that uses a vehicle with 3 omnidirectional wheels to navigate. This is an important inspiration for us, as we decided that the three wheel vehicle was the good way to implement this. Since we want to make our own BB8 in terms of speed, wobbling, etc. We will not use the list of materials he provided, and this is also not goal of this course to copy other's projects.

His website:

<https://makezine.com/projects/build-full-size-working-bb-8-using-beaglebone/>

3. User Stories

As a **user**, I want the BB8 to be able to move smoothly in all directions, so that it seems like a real free agent.

As a **user**, I want to be able to wirelessly control the BB8 with my phone/laptop, with a free interface such as a webapp (or even an application) that is easy and intuitive to use, so that i don't need to carry a special controller with me all the time.

As an **EPFL visitor**, I want the BB8 to be able to recognize objects/faces so that it can come and welcome me, follow me, and play with me.

As a **software developer**, I want the BB8 code to be well organized and documented, so that i can quickly modify it, easily understand it and clearly explain it.

As an **EPFL teacher**, I want the BB8 to be optimized in terms of cost, weight and size so that it is affordable and storable.

4. Product Management & Afterlife

Potential follow-up work

- Autonomous navigation
- Person-following via the camera and onboard vision algorithms
- Gesture-based control replacing the gamepad

Beyond EPFL

- Demo robot for public outreach events (the form factor is immediately recognisable)
- Research platform for testing holonomic control algorithms
- Reception or guidance robot in museums and science exhibitions
- Hardware base for student robotics competitions

5. Cost Estimate

Part	Assembly	Subassembly	Qty	Price (unit)	Price (total)	Link
XIAO ESP32S3	camera + chip	Head	1	CHF 11.77	CHF 11.77	Digikey
BNO055 9-Axis Sensor	Accelerometer + Gyroscope	Body	1	CHF 19.90	CHF 19.90	Bastelgarage
DC 12V 251RPM 18Kg*cm	Motors	Body	3	CHF 35.90	CHF 107.70	Bastelgarage
Double BTS7960 43A H-Bridge	Driver	Body	3	CHF 7.00	CHF 21.00	parts in stock cs358
GensAce 5Ah 11.1V 60C 3S1P	Body battery	Body	1	CHF 32.70	CHF 32.70	Galaxus
3S 60A 12V LiPo protection board	Battery protection board	Body	1	CHF 15.00	CHF 15.00	parts in stock cs358
XT90 plugs	XT90 Connector male and female	Body	2	CHF 6.9	CHF13.80	Bastelgarage
12 AWG cable (4mm ²)	–	Body	1	CHF 5.40	CHF 5.40	Conrad
xh2.54 male plug	–	Body	1	CHF 0.20	CHF 0.20	Mouser
7-20V to 5V 3A Buck Converter	Buck Converter	Body	1	CHF 10.90	CHF 10.90	Bastelgarage
Magnetic disc Ø20mm h5mm N42	Magnet	Body, Head	6	CHF 2.00	CHF 12.00	Supermagnete
Li-Ion NCR18650B 3.7V 3200mAh	Head battery	Head	1	CHF 8.90	CHF 8.90	Bastelgarage
1x18650 Battery Shield 5V 3A	Head battery shield	Head	1	CHF 9.90	CHF 9.90	Bastelgarage
Balls Ø14mm	Ball support	Head	6	CHF 3.30	CHF 19.80	123roulement
Wemos D1 R32	Microcontroller	Body	1	CHF 0.00	CHF 0.00	parts in stock cs358
SG90	Head Servo	Head	1	CHF 3.00	CHF 3.00	parts in stock cs358
Collier de selle 28.6mm	Seat Clamp	Body	1	CHF 6.90	CHF 6.90	Decathlon
Magnetic disc Ø2mm h2mm N42	Magnet (closure)	Body	60	CHF 0.18	CHF 10.80	Supermagnete
TOTAL					CHF309.47	

Table 1: Bill of Materials

6. Technical Feasibility and Key Challenges

WiFi and/or Bluetooth communication with the ESP32

The robot relies on wireless communication (WiFi and/or Bluetooth) with an external device (controller). The ESP32 supports both protocols, but can always cause unpredictable latency if not treated correctly and range may also become an issue in environments with interference. Since the robot's movement and camera feed both depend on this link, any communication failure directly impacts the core functionality of the project.

Solutions: We will start by building a dedicated communication prototype in the first weeks we receive the components: two ESP32s exchanging movement commands in a loop, where we measure latency and packet loss under realistic conditions. We will pick one

primary protocol (most likely WiFi UDP for its low latency) and avoid running both WiFi and Bluetooth simultaneously unless strictly necessary.

Different voltage requirements across components

The system includes components with very different power needs: DC motors typically run at 6–12V, while the ESP32 operates at 3.3V logic (5V input), and the camera module has its own requirements. Managing multiple voltage rails with converters introduces risks of noise on logic lines, brown-outs during motor startup, and incorrect wiring that could damage components.

Solutions: Before wiring anything, we will draw a complete power distribution schematic listing every component, its required voltage, and its peak current draw. We will then build a power prototype early on (wiring all consumers (ESP32, motors, camera) on the real batteries with the planned voltage converters) and verify that the ESP32 does not reset when the motors start. We will add decoupling capacitors on every power line.

Magnetic coupling to hold the head

The head of the BB-8 is held onto the body using magnets. The coupling must be strong enough to keep the head attached during acceleration and direction changes, yet free enough to allow smooth rotation in any direction. Finding the right magnet strength, size, and placement is a physical problem that is difficult to reason about purely on paper. Additionally, strong magnets placed near the accelerometer or compass could interfere with sensor readings, introducing errors in the stabilization system.

Solutions: We made several calculations on the head weight and the required magnets strength that should guarantee good results.

Sphere stability and wobble

The internal 3-wheeled chassis pushes the sphere from the inside using friction. If the weight distribution is uneven or the wheels are poorly positioned, the sphere will wobble, spin in place, or veer unpredictably. More critically, sudden acceleration or braking causes the internal chassis to shift momentum abruptly, which can throw the entire system off balance and send the sphere in an unintended direction. Controlling smooth, stable motion therefore requires both a well-balanced mechanical design and a reliable closed-loop control system using the accelerometer, and a failure in either layer will make the robot unusable.

Solutions: To address sphere stability, we will place the heaviest components as low as possible on the chassis to keep the center of mass close to the ground, which passively reduces wobble. We will then use an accelerometer combined with a software control loop to continuously monitor and adjust the motor commands, compensating for unwanted movements in real time.

Camera performance and feature support

The camera mounted in the head needs to deliver sufficient image quality to support features like face detection. An ESP32-CAM module can handle basic streaming at low frame rates, but real-time face detection using deep learning models requires significantly more processing power. If the camera is connected to the same ESP32 handling motor

control, it may saturate the CPU and degrade both systems. The risk is not just whether the camera works, but whether it can run the intended features at an acceptable speed within the constraints of embedded hardware.

Solutions: We will prototype the camera system completely independently, getting face detection running on a standalone device before any robot integration. We will use a dedicated processor for vision (a separate ESP32) so it never competes with motor control for CPU time. If on-board inference is too slow for real-time detection, we will offload the computation to an external laptop over WiFi: the robot streams video out, the laptop runs the detection model and sends back high-level commands.

Keeping the sphere small enough

All internal components (the chassis, motors, batteries, ESP32, wiring, and camera) must fit inside the sphere. The available volume is more constrained than it appears in early sketches. Cables, connectors, and the need for physical access (e.g., to recharge batteries or debug hardware) add hidden bulk. If the sphere is too small, the chassis cannot be built as designed; if it is made larger to compensate, the robot loses its aesthetic and may become harder to control mechanically.

Solutions: This has already been done on the cad. We have left some margin for mistake.

Insufficient motor torque

The motors must move the entire robot (chassis, batteries, and all electronics) by pushing against the inside of the sphere. If the motors lack sufficient torque, the robot will struggle to accelerate, climb slight inclines, or recover from perturbations. This risk is compounded by the indirect drive mechanism (friction against the sphere wall), which is inherently less efficient than direct drive. Motor selection must account for the total system weight, the friction coefficient between wheels and sphere, and the expected loads during normal operation.

Solutions: We already estimated the required torque before ordering motors: calculating the total system weight and the expected friction force between wheels and sphere, and selecting motors that exceed this with clear margin. We will then build the chassis at its intended full weight and verify on a flat surface that the motors can accelerate and stop without stalling. If this isn't enough, we will add/create some kind of gear to adjust the power.

Omni-wheel positioning causing excess friction or mechanical resistance

The three omni-wheels must be positioned at angles that allow movement in any direction while maintaining consistent contact with the inside of the sphere. If the geometry is wrong (wrong angles, wrong contact pressure, or misaligned axles) the wheels will generate lateral friction forces that resist motion rather than enabling it. This is a subtle mechanical design problem: small errors in wheel placement can create binding, uneven wear, or loss of directional control. It is also difficult to diagnose once the chassis is assembled inside the sphere.

Solutions: We calculated the angles with the CAD and will print our own omni-wheels so we will be able to adjust them to our specific requirements.

Sphere structural integrity and cracking

The sphere is a critical structural component that must withstand continuous mechanical stress from internal forces, impacts with the environment, and repeated opening/closing if the system is accessible. Depending on the material (e.g., PLA, PETG, polycarbonate), the sphere may be prone to cracking, especially along layer lines if 3D printed. Small defects or stress concentrations (such as holes for ventilation or assembly seams) can propagate into larger fractures over time. A cracked sphere not only compromises the aesthetic but can completely break the locomotion system, as the internal traction relies on a smooth and rigid surface.

Solutions: We made several tests with smaller 3-D printed spheres and talked with Sebastien who told us the integrity shouldn't be a problem.

Imperfect sphere geometry

If the sphere is not perfectly round (due to manufacturing tolerances, deformation, poor assembly, or uneven material thickness), the robot may experience irregular motion. Even small deviations in curvature can change how the omni-wheels contact the surface, creating vibration, wobble, or inconsistent rolling behavior. Since the entire locomotion concept depends on predictable contact between the wheels and the sphere, geometric imperfections can significantly degrade performance.

Solutions: We will validate the sphere's roundness before assembling the chassis inside it: rolling it on a flat surface and checking for wobble, and measuring the internal diameter at multiple points. Once 3-D printed, we will sand and finish the internal surface to reduce irregularities.

7. Gantt Chart

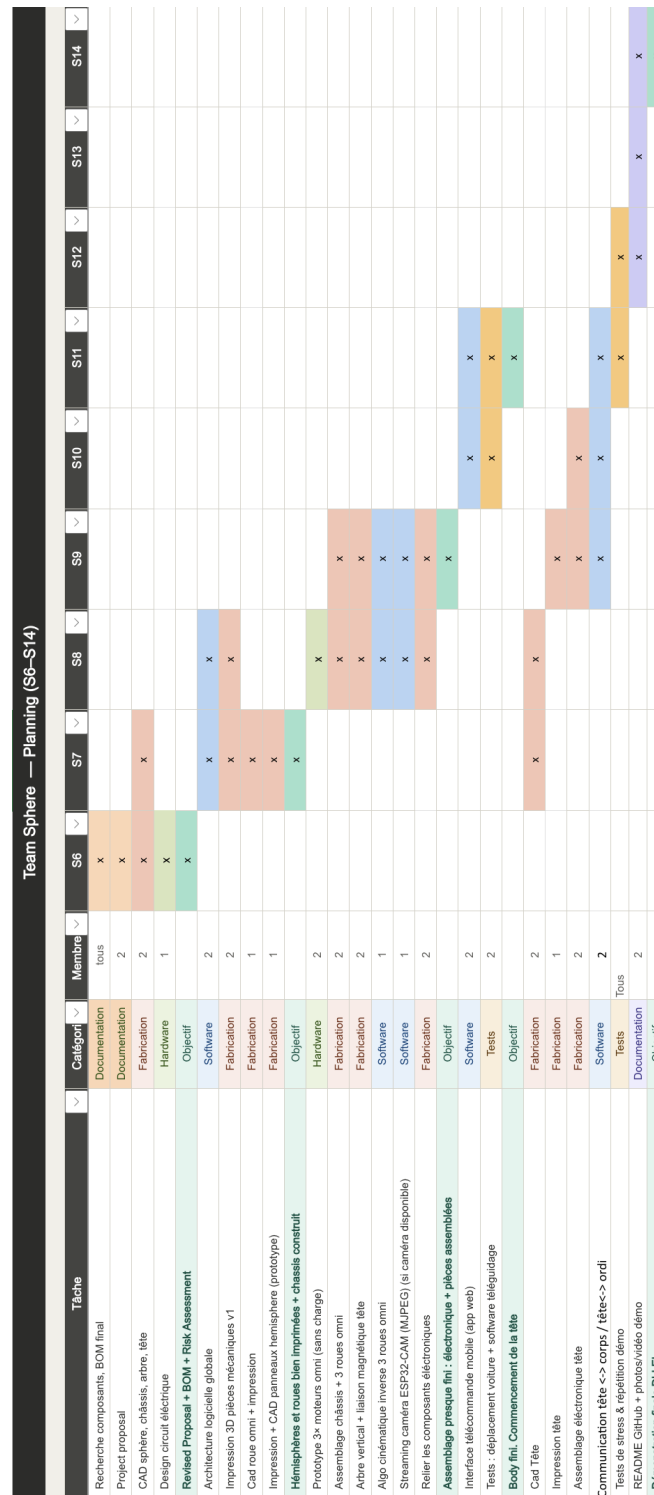


Figure 5: Gantt

8. CAD



Figure 6: CAD1

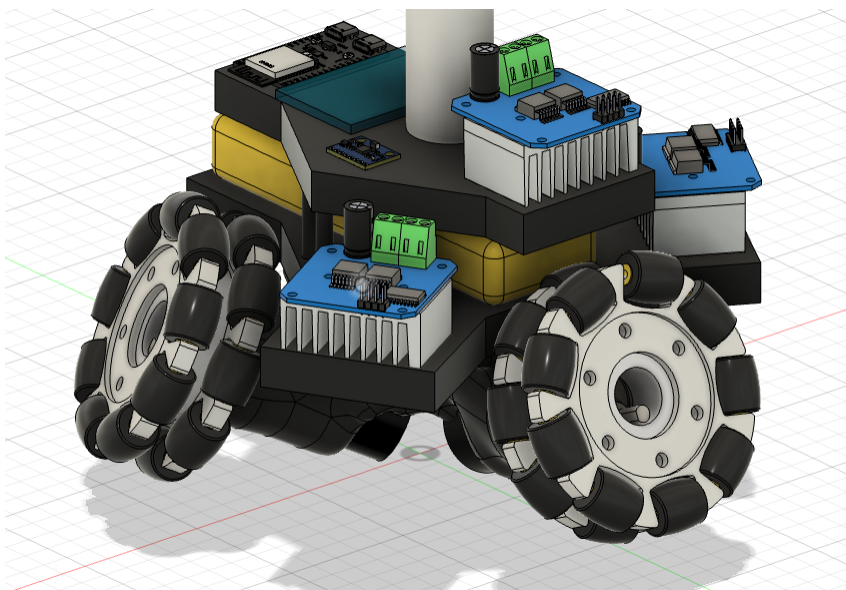


Figure 7: CAD2

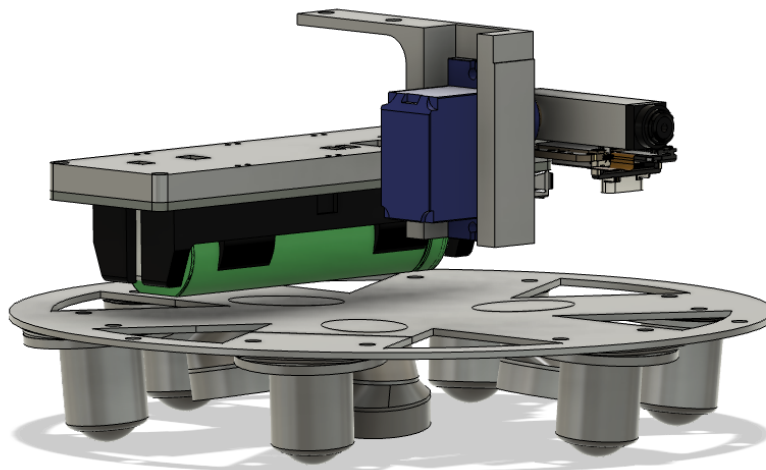


Figure 8: CAD3

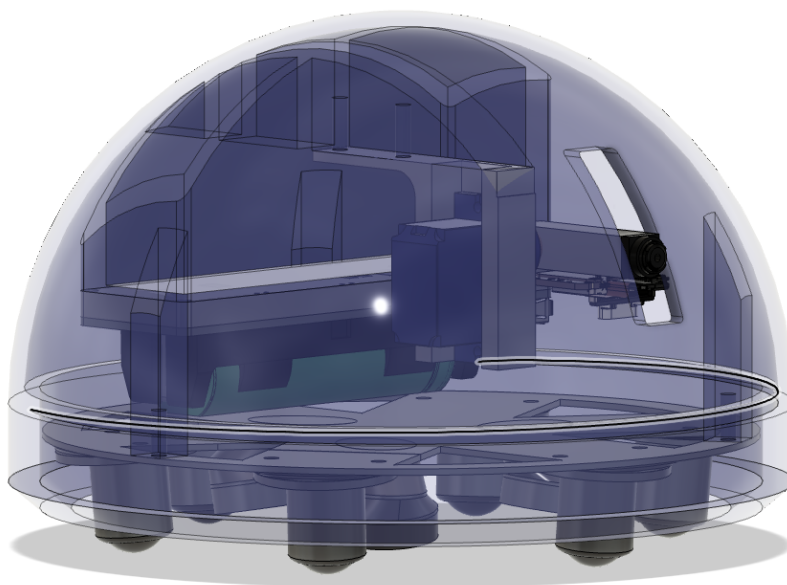


Figure 9: CAD4

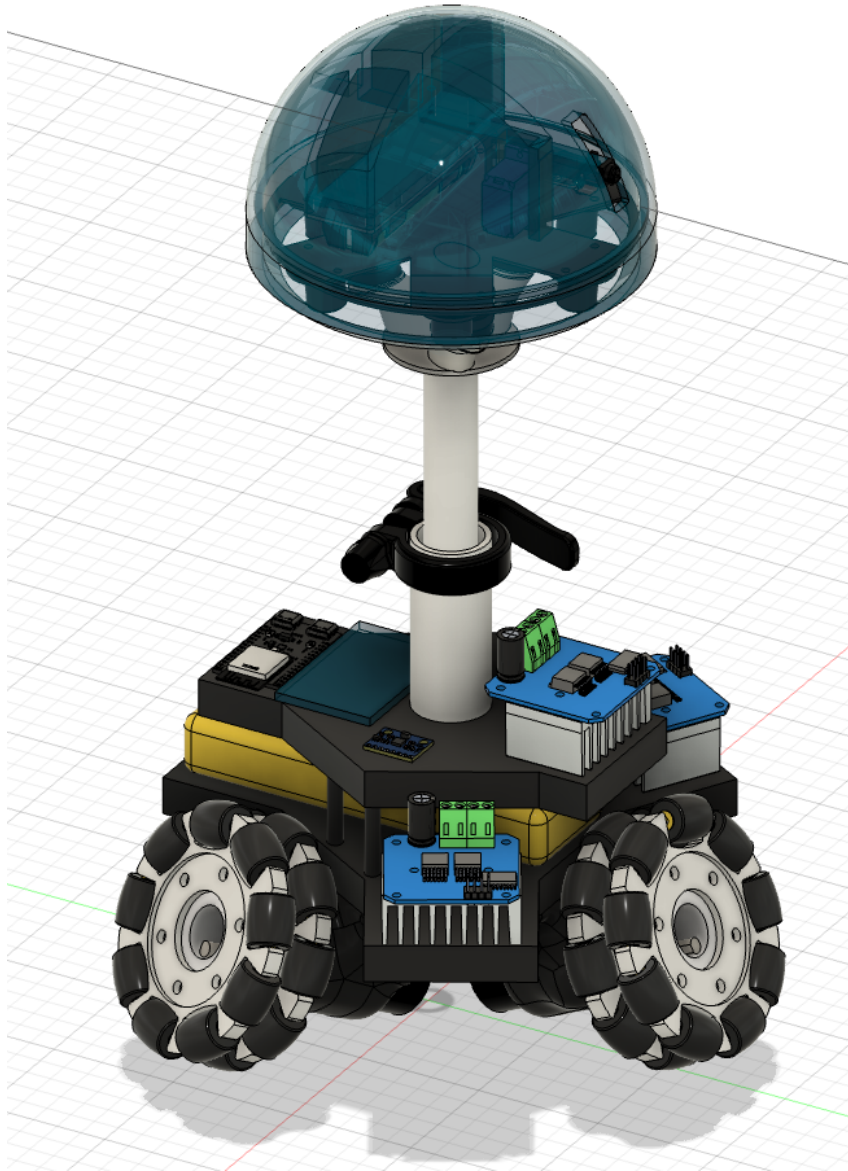


Figure 10: CAD5

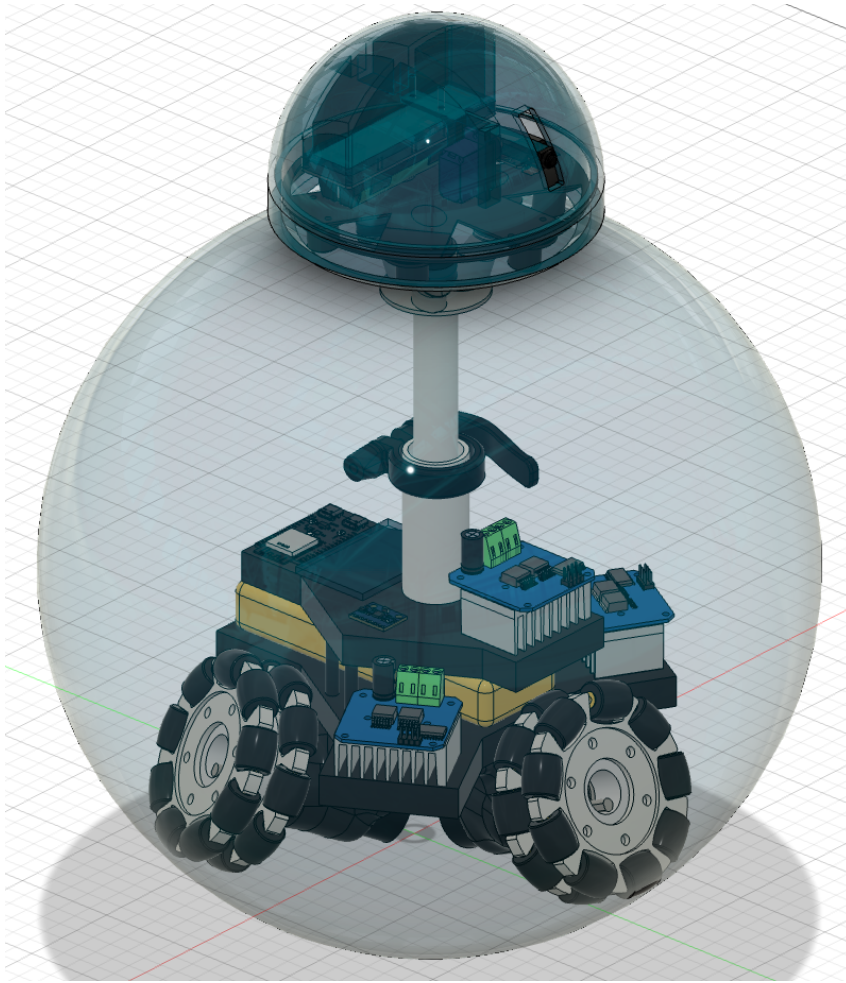


Figure 11: CAD6